



Bachelor's Thesis

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Degree Programme
Systems Engineering

Major
Infotronics

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Probes for DOTS (Distributed OT Test Suite)

Graduate

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Objective

The objective of this work is to develop and implement an embedded probe capable of performing various test operations within OT environments, collecting their results, and communicating them to a central coordinator.

Methods | Experiences | Results

Operational Technology (OT) environments require numerous functional and security-related tests to be defined, repeated, and logged whenever a system is modified. Preparing and conducting these tests can be complex, as operations must be performed at different points of the system and their effects observed. This work focuses on the development of configurable probes in the form of embedded systems for distributed OT testing.

The probe is a configurable and remotely controlled device capable of performing various commands when deployed in an OT environment. It connects to a central coordinator and advertises its capabilities and configurable parameters, allowing it to be adapted to different test environments. The coordinator can then send commands or configuration updates, while the probe executes the requested commands and returns their results. Communication relies on MQTT, while local logging provides traceability of executed operations.

The solution was validated on a physical test bench composed of three probes and one Device Under Test. A test suite was conducted to evaluate the probe's operation and interoperability with the DOTS coordinator.